

ROLLER COASTER B

Team Checklist – 2027

Rank: _____
Tier: _____
Final Score: _____

Team Number: _____ School & Team Name: _____

Student Names: _____

IMPOUND

1. T F 4.a. Team impounds one Roller Coaster, and any ball(s)/sphere(s), tools, spare parts, and data/notes.

CHECK-IN AND SAFETY

T F 2.c. Participants properly wear Class B eye protection at all times.

T F 2.e. There are no unresolved safety issues as deemed by the Event Supervisor.

2. T F **BOTH OF THE ABOVE SAFETY PARAMETERS ARE MET (If False, do not run and give Participation Points)**

Construction Parameters:

T F 3.b. At all times within the competition, the device, excluding the ball, is no larger than 30.0 cm wide x 80.0 cm long x 60.0 cm high sitting flat.

3. _____ 5.d. **Device Width:** Must be no larger than 30.0 cm, measured to the nearest 0.1 centimeter.

4. _____ 5.d. **Device Length:** Must be no larger than 80.0 cm, measured to the nearest 0.1 centimeter.

5. _____ 5.d. **Device Height:** Must be no larger than 60.0 cm, measured to the nearest 0.1 centimeter.

T F 3.c. The ball is at least 1 cm in diameter and remains visible at all times.

T F 3.f. There is exactly one clearly labeled Start Line and one clearly labeled Finish Line running perpendicular to the direction of the ball travel on the track designated before the Target Time is released.

T F 3.g. The relative positions of the Start Line and Finish Line do not change after impound.

T F 3.h. The device includes a mechanism that safely stops the ball after it crosses the Finish Line

T F 3.i. Magnets, electrical, and electronic devices are not used for any part of the Roller Coaster.

T F 2.f. Team can answer questions on design, construction, and operation of the device per the Building Policy found on www.soinc.org

1ST RUN 2ND RUN ← EVENT TIME (8 MINUTES FOR UP TO 2 SCOREABLE RUNS)

Construction Parameters:

T F T F ALL CHECK-IN CONSTRUCTION PARAMETERS ARE MET (Recheck construction parameters if necessary)

T F T F 3.d. The ball is held in the ready-to-run position by an unsharpened #2 pencil held only in the participant's hand on the Start Line of the device. The ball is released when the participant removes the pencil from the track.

T F T F 3.e. The ball travels using only its own gravitational potential energy available at the ready-to-run position with the exception of the simple machines required for the Elevators. No energy is added by use of stored potential energy.

6. T F 12. T F **ALL CONSTRUCTION PARAMETERS ARE MET. If F, Tier 2.**

Competition Parameters:

T F T F 4.b. Once participants enter the event area to compete, they do not leave or receive outside assistance, materials, or communication until they are finished competing and have left the event area.

T F T F 4.g. Prior to conducting a scoreable run, participants place a #2 pencil on the Start Line and the Ball is placed completely behind the Start Line.

T F T F 4.h. Participants do not touch the device during a scoreable run.

7. T F 13. T F **ALL COMPETITION PARAMETERS ARE MET. If F, Tier 2.**

8. T F 14. T F 4.k. **Finish Line:** The Ball crosses the Finish Line. (If the ball fails to cross the Finish Line on a scoreable run, record the time of the run when it stopped or left the device to be used as the final Tie Breaker (if needed) in the Run Time box)

9. _____ 15. _____ 4.j. **Run Time:** Begins on the word "Go" when the pencil is removed and ends when any of the following happens:
1) The ball completely crosses the Finish Line
2) Twice the Target Time has elapsed since the word "Go".
3) The ball travels outside the boundary of the device.
4) The ball stops moving, and does not begin moving again within 3 seconds.

10. _____ 16. _____ 5.f. **(Optional) Elevator Bonus (Write 0, 1, or 2):** Number of distinct and clearly labeled Elevators included. Each type of elevator only earns points once. **(Only fill out if Elevator is successfully used as defined in 3.j.)**
3.j. **Type 1:** A first-class lever combined with a wheel and axle with an IMA of <1 raises the ball vertically at least 10 cm.
Type 2: A second-class lever combined with a pulley with an IMA of <1 raises the ball vertically at least 10 cm.

11. _____ 17. _____ 5.g. **(Optional) Loop Bonus:** The height of the Loop (cm) measured vertically from the Point of Intersection to the highest inner point of the Loop. **(Only fill out if ball successfully completes the Loop and fulfills all requirements in 3.k.)**

3.k. 1) A Loop is included in the device.
2) The track is not fully enclosed at any at any point in the Loop.
3) The measurement from the POI to the highest inside point of the Loop is clearly labeled
4) The two sections of track at the Point of Intersection must touch and have a clearly labeled point.
5) The ball must remain in contact with the track as it traverses the Loop. When the ball is in contact with the highest point of the Loop, the ball must be completely unsupported from below.

18. T F General Rule: The team is disqualified. (Notify the team and their coach as soon as possible.)